Draft-Strategic Data Masking & Classification Design Document

Document Control

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Name	Role	Purpose	Version	Date	Status

Key Revision History

Version	Date	Description	Author(s)	Status
0.1	24/02/2026	First Draft	Naresh Kala	Draft

Summary

The architecture ensures that classification metadata extracted from Alation flows through a governed dbt pipeline into the Snowflake Governance Database, where it drives automated tag application and masking policy enforcement across all data product tables.

Objectives

- Automate ingestion of classification metadata from Alation
- Eliminate manual classification entry
- Standardize tagging and masking policies
- Enable Auditable governance

Current State (As-Is)

The following components have been confirmed as existing and available for this initiative

- Manual entries in CLASSIFICATION_DATA
- dbt macro applies tags
- Governance Database .
- Masking policies applied via tags

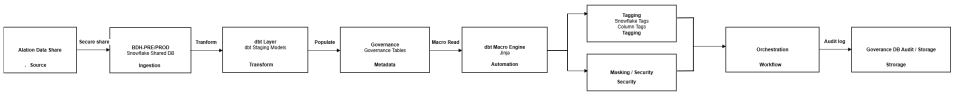
Target State (To-Be)

The following components are identified as gaps requiring design, development, and delivery:

- dbt Project: A new dbt project targeting GOVERNANCE_DB, with Source Staging Classification model layers.
- dbt Macros: Three macros — apply_classification_tags, apply_masking_policies, apply_object_tags — require design and implementation.
- CI/CD Pipeline: GitHub repository, branching strategy, environment promotion, and automated testing pipeline.
- Airflow Orchestration: Astro workspace provisioning, DAG development for end-to-end pipeline scheduling.
- RBAC Implementation: Grants across CLASSIFICATION, TAGGING, MASKING, and LOGGING schemas .
- Test Strategy: Unit tests, macro validation, end-to-end masking enforcement tests.

Architecture Overview

Alation Snowflake DB BDH-Alation snowflake Shared DB dbt Staging Governance Tables dbt Macro Tags Masking Policies Orchestration Governance DB



Core Architecture Flow

Layer	Component	Responsibility
Source	Alation snowflake Shared DB	Source of classification metadata (tags, sensitivity, column assignments)
Staging	dbt Staging Models	Extract, validate and standardize Alation share data

Classification	dbt Classification Models	Apply business rules; write to CLASSIFICATION_DATA table
Tagging	dbt Macros (apply_tags)	Execute ALTER TABLE to apply Snowflake tags from classification data
Masking	dbt Macros (apply_masking)	Assign masking policies to tagged columns in Snowflake
Orchestration	Airflow (Astro)	Schedule and sequence the end-to-end pipeline
Governance DB	Snowflake – GOVERNANCE_DB	Target database hosting CLASSIFICATION, TAGGING, MASKING, LOGGING schemas

Data Classification Requirements

Alation Source Metadata(TBD)

The following fields are extracted from the Alation Snowflake Data Share as source inputs for classification, tagging, and masking. All source references must be validated against the approved STM document before staging development commences.

Source Field	Description	Validation Rule
tag_name	Classification label applied in Alation	Must match allowed list: PII, PCI, CONFIDENTIAL, PUBLIC, SENSITIVITY_LEVEL
tag_value	Value associated with the tag	Must match sensitivity levels: High, Medium, Low, None
object_type	Type of Snowflake object (TABLE /COLUMN)	Must be TABLE or COLUMN
object_schema	Schema containing the object	Must exist in GOVERNANCE_DB
object_table	Table name the tag applies to	Must resolve to an existing Snowflake table
object_column	Column name (null for table-level tags)	Must exist in the referenced table if object_type = COLUMN
assigned_by	Alation user who assigned the classification	Free text; must not be null
assigned_date	Timestamp of classification assignment	Must be a valid timestamp; used for incremental load logic

Data Product Scope(TBD)

A complete data product table inventory must be agreed with the Data Governance team and documented in the confluence page . The table below represents the classification category framework.

Classification	Sensitivity	Masking Policy	Masking Approach	Example Fields
PII	High	MASK_PII_HIGH	SHA-256 Hashing	full_name, email, date_of_birth, nhi_number
PCI	High	MASK_PCI_HIGH	Full Redaction	card_number, cvv, expiry_date
Confidential	Medium	MASK_CONF_MED	Partial Masking	salary, account_balance, internal_id
Public	None	None	No Masking	product_name, region_code, public_metrics

Source-to-Target Mapping (STM)

STM Overview

The Source-to-Target Mapping document defines the complete column-level lineage from the Alation Data Share through to the CLASSIFICATION_DATA table in GOVERNANCE_DB. The STM is the governing reference for all dbt model development and must be version-controlled alongside the dbt project.

Column-Level Mapping — CLASSIFICATION_DATA(TBD)

Target Column	Source Field	Transformation	Data Type	Nullable
CLASSIFICATION_ID	Derived	HASH(object_schema||object_table||object_column||tag_name)	VARCHAR(64)	No
OBJECT_SCHEMA	object_schema	UPPER(TRIM(object_schema))	VARCHAR(255)	No
OBJECT_TABLE	object_table	UPPER(TRIM(object_table))	VARCHAR(255)	No
OBJECT_COLUMN	object_column	UPPER(TRIM(object_column))	VARCHAR(255)	Yes
OBJECT_TYPE	object_type	UPPER(object_type)	VARCHAR(10)	No
CLASSIFICATION_TYPE	tag_name	UPPER(TRIM(tag_name))	VARCHAR(50)	No
SENSITIVITY_LEVEL	tag_value	UPPER(TRIM(tag_value))	VARCHAR(20)	No

MASKING_POLICY_NAME	Derived	CASE classification_type WHEN PII THEN MASK_PII_HIGH ...	VARCHAR(100)	Yes
ASSIGNED_BY	assigned_by	Direct copy	VARCHAR(255)	Yes
ASSIGNED_DATE	assigned_date	CAST(assigned_date AS TIMESTAMP_NTZ)	TIMESTAMP_NTZ	Yes
RECORD_CREATED_AT	System	CURRENT_TIMESTAMP()	TIMESTAMP_NTZ	No
RECORD_UPDATED_AT	System	CURRENT_TIMESTAMP() on merge	TIMESTAMP_NTZ	No
IS_ACTIVE	Derived	TRUE unless superseded by newer record	BOOLEAN	No

Merge Key Strategy

Primary merge key for incremental loads: (OBJECT_SCHEMA, OBJECT_TABLE, OBJECT_COLUMN, CLASSIFICATION_TYPE). This composite key uniquely identifies each classification assignment at the column level. For table-level tags where OBJECT_COLUMN is null, OBJECT_COLUMN should be stored as the literal string '__TABLE__' to enable the merge key to function correctly.

Incremental Processing

Strategy

- Use record_updated_at timestamp
- Hash-based change detection

Handling Changes

- New records Insert
- Updated Update
- Removed Optional delete/flag

Masking Policy Name Mapping(TBD)

Classification Code	Sensitivity Level	Masking Policy Name	Notes
Unclassified	High	~~MASK_PII_HIGH~~	Hashing approach – SHA-256
High Confidential	High	~~MASK_PCI_HIGH~~	Full redaction
Confidential	Medium	~~MASK_CONF_MED~~	Partial masking – first 2 chars + asterisks
Private	Low	~~MASK_CONF_LOW~~	Partial masking – first 4 chars visible
Public	None	~~NULL – no policy~~	No masking applied

dbt Architecture Design

Project Structure

A new dbt project will be created targeting GOVERNANCE_DB. The project sources data from the Alation Snowflake Data Share (read-only) and writes transformed classification data into the CLASSIFICATION, TAGGING, MASKING, and LOGGING schemas.

Model Layer Design

Layer	Model	Materialisation	Target Schema	Description
Source	TBD	Source (external)	ALATION_SHARE	Read-only references to Alation Data Share tables
Staging	TBD	View	CLASSIFICATION	SELECT + validate + standardise from Alation share
Staging	TBD	View	CLASSIFICATION	Extract sensitivity level assignments
Classification	TBD	Incremental Table	CLASSIFICATION	Core model — final classification output with merge logic
Tagging	TBD	Incremental Table	TAGGING	Records of Snowflake tag assignments to apply
Masking	TBD	Incremental Table	MASKING	Records of masking policy assignments to apply
Logging	TBD	Incremental Table	LOGGING	Execution logs, row counts, error capture

dbt Project Configuration (dbt_project.yml)

The dbt_project.yml must define the following environment-specific targets: dev (developer Snowflake schema), syst (system integration), ppte (pre-production), and prod (production). Database and schema targets must be parameterised via environment variables to support CI/CD promotion without config changes.

Macro Design

Macro 1: apply_classification_tags

Reads from CLASSIFICATION_DATA and generates ALTER TABLE / ALTER COLUMN statements to apply Snowflake tag objects. The macro iterates each row in the classification output and constructs the appropriate SQL. It is idempotent — safe to re-run without creating duplicate tag assignments. Supports all classification types (PII, PCI, CONFIDENTIAL, PUBLIC).

Classification Tag (TBD)

- PII TAG_PII
- CONFIDENTIAL TAG_CONFIDENTIAL

Macro 2: apply_masking_policies

Reads from MASKING_ASSIGNMENTS and generates the sequence of SQL statements required to apply Snowflake masking policies. For each record, the macro first issues an UNSET MASKING POLICY statement to remove any existing policy before applying the new one, preventing policy conflicts. Designed to be safely re-executed in all environments.

Sensitivity Masking (TDB)

- HIGH FULL_MASK
- MEDIUM PARTIAL_MASK
- LOW NONE

Macro 3: apply_object_tags

Reads tag assignments and applies object-level tags at both table and column granularity using ALTER TABLE SET TAG and ALTER TABLE ALTER COLUMN SET TAG syntax respectively. The macro generates the required SQL dynamically and is safe to run multiple times without causing errors or duplicate changes.

Hashing — BK / PK / FK Implementation

MASK_SHA256_HASH policy created apply on Key flag columns in CLASSIFICATION_DATA populated . BK/PK/FK tags provided by Data Governance.

Logging & Error Handling

- All macro executions write audit records to GOVERNANCE_DB.LOGGING.AUDIT_LOG, capturing: run timestamp, macro name, row count processed, success/failure status, and error message if applicable.
- dbt model run results are captured via dbt's built-in results object and persisted to the audit log table.
- Failed records are written to a LOGGING.ERROR_LOG table with the full record and error reason for investigation.
- Alerting on pipeline failures is triggered via Airflow task failure callbacks.

Incremental Processing

Strategy

- Use record_updated timestamp
- Hash-based change detection

Handling Changes

- New records Insert
- Updated Update
- Removed Optional delete/flag

Macro Responsibilities

- Read classification data
- Compare existing tags
- Apply delta changes
- Log actions

Source Setup & Alation Share Validation

Before staging model development, the following steps must be completed to ensure source integrity:

- Configure dbt sources to reference the Alation Data Share database in read-only mode.
- Confirm the dbt service account has SELECT privileges on all required Alation share tables.
- Profile all Alation share tables to validate structure, data types, and record counts against the approved STM.
- Validate that tag_name values in the share are restricted to the approved classification list: PII, PCI, CONFIDENTIAL, PUBLIC, SENSITIVITY_LEVEL.

DevOps Workflow & CI/CD

Branching Strategy

Branch	Purpose	Merge Target	PR Required
feature/*	Individual development tasks	develop	Yes — peer review
develop	Integration branch for testing	prod	Yes — tech lead approval
Prod	Production-ready code	—	Yes — governance sign-off

Orchestration — Airflow (Astro)

Pipeline DAG Design

The end-to-end classification pipeline will be orchestrated via Airflow running on the Astro platform. The Astro Airflow DAG that executes the full pipeline: Alation share validation, staging, classification, tagging, masking, and post-validation. Supports schedule and ad-hoc manual trigger. Passes time-based parameters to dbt at runtime.

Work flow

- Refresh metadata
- Run dbt staging
- Update classification table
- Apply tags
- Validate

Role-Based Access Control (RBAC)

RBAC Matrix(TBD)

Role	PII Column	HC Column (default)	CONF Column	PUBLIC Column	TRANSFORM Own Schema	Another Dev's TRANSFORM Schema
HC Developer (zone-specific)	Plaintext	Plaintext	Plaintext	Plaintext	Plaintext — own only	DENIED
CONFIDENTIAL Analyst	SHA-256 Hash	SHA-256 Hash	Plaintext	Plaintext	No access	No access
PUBLIC Analyst	NULL / MASKED	NULL / MASKED	Partial (2 chars)	Plaintext	No access	No access
Airflow Service Role	Plaintext	Plaintext	Plaintext	Plaintext	No access	No access
dbt Service Role	Plaintext	Plaintext	Plaintext	Plaintext	No access	No access

Testing

Test Scope

Test Type	Scope
dbt Schema Tests	All models — not null, unique, accepted values, referential integrity
Macro Unit Tests	Validate generated SQL for apply_classification_tags, apply_masking_policies, apply_object_tags
Integration Tests	Staging Classification model chain; data flows correctly end-to-end
Row Count Reconciliation	CLASSIFICATION_DATA row count vs Alation share source record count
Tag Verification	Confirm Snowflake tags applied correctly to all in-scope tables and columns
Masking Verification	Confirm masking policies applied; test role-based masking behaviour
Hashing Validation	Verify SHA-256 hashing applied correctly for PII fields
End-to-End Test	Full pipeline: Alation share staging classification tag application masking enforcement Governance DB

Conclusion

This solution provides a scalable, automated, and governance-driven framework leveraging Alation Data Share and Snowflake capabilities to enforce consistent data protection policies across the organization.